

Liyuan Liu

PRINCIPAL RESEARCHER @ MSR

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“We are trying to prove ourselves wrong as quickly as possible, because only in that way can we find progress. — Richard Feynman”

SUMMARY

Led pre-training infrastructure and recipe design for Phi-3.5-MoE, with our methodology adopted by Qwen3-MoE. Advanced key LLM technologies including initialization, learning rate warmup, KV cache compression, and gradient estimation.

EXPERIENCE

Microsoft Research

PRINCIPAL RESEARCHER

WA, USA

Feb 2025 – PRESENT

Microsoft Research

SENIOR RESEARCHER

WA, USA

Feb 2022 – Feb 2025

EDUCATION

University of Illinois Urbana-Champaign

PH.D. IN COMPUTER SCIENCE

Illinois, USA

September 2016 – May, 2024

- Thesis: Understanding the mechanism of pretraining stabilization heuristics: A variance-oriented perspective.
- Advisor: Prof. Jiawei Han.

University of Science and Technology of China

B.E. IN COMPUTER SCIENCE

Hefei, China

September 2012 – June 2016

- Advisor: Prof. Linli Xu. Graduated *summa cum laude*.

SELECTED HONORS

ICLR 2024 Outstanding Paper Honorable Mention

2024

- Shortlisted as 1 of 16 papers among 4922 submissions.

Winner of the Topcoder Arabic NER Challenge

2018

- Ranked 1st among 137 registrants and 220 submissions.

Google Excellent Scholarship

2015

- Only 58 graduate and undergraduate recipients nationwide.

SELECTED PROJECTS

RAdam	[included in PyTorch] [2600+ stars & 300+ forks] if warmup is the answer, what is the question?
Phi-3.5-MoE	[partially adopted by Qwen3-MoE] [500+ stars] lead MoE pre-training algorithm and infra development
LM-LSTM-CRF	[800+ stars & 200+ forks] first character-level LM -based representation learning method
Transformer Clinic	Transformer initialization for training stabilization
LD-Net	first pre-trained language model inference acceleration method
Flash-RL	modern framework implicitly brings RL to off-policy and can be accelerated via quantization
Torch-Scope	[80k+ downloads] toolkit for automatic training logging and visualization

SCHOLARSHIPS & AWARDS

Yee Fellowship Award	2020
Data Mining Group Research Award	2019, 2018, 2017
3th Place at WSDM Cup 2017 (Vandalism Detection Task)	2017
Guo Moruo Scholarship	2015
Samsung Scholarship	2015
Tencent Innovative Scholarship	2014
Seagate Scholarship	2013
Scholarship for Outstanding Students	2012, 2013

INTERNSHIPS

Deep Learning Group @ MICROSOFT RESEARCH REDMOND

PART-TIME RESEARCHER, ADVISOR: DR. XIAODONG LIU & DR. JIANFENG GAO

IL, USA

May 2021 – December 2022

AI Lab @ TENCENT

RESEARCH INTERN, ADVISOR: DR. JIANSHU CHEN & DR. LIWEI WANG

WA, USA

May 2020 – August 2020

Deep Learning Group @ MICROSOFT RESEARCH REDMOND

PART-TIME RESEARCHER, ADVISOR: DR. XIAODONG LIU & DR. JIANFENG GAO

IL, USA

January 2020 – May 2020

D365 AI & MSR @ MICROSOFT

RESEARCH INTERN, ADVISOR: DR. WEIZHU CHEN & DR. JIANFENG GAO

WA, USA

May 2019 – August 2019

AllenNLP @ AI2

RESEARCH INTERN, ADVISOR: DR. MATTHEW E. PETERS

WA, USA

May 2018 – August 2018

Social Computing Group @ MICROSOFT RESEARCH ASIA

UNDERGRADUATE RESEARCH INTERN, ADVISOR: DR. RUIHUA SONG

Beijing, China

July 2015 – May 2016

PUBLICATIONS

SUMMARY

My research focuses on heuristic tricks and their mechanisms that empower Large Language Modeling (LLM) training and inference, including initialization, gradient estimation, lr warmup, and kv cache compression, has been **cited 7,600+ times**, received about **4,000+ stars** and **600+ forks** on Github, been awarded a **Outstanding Paper Honorable Mention** in ICLR2024 and a **Grand Prize** in the Topcoder Named Entity Recognition Challenge, been implemented to popular frameworks like PyTorch and OpenRLHF, been adopted to training recipes for popular models like Qwen3-MoE, placed first in various benchmarks (e.g., WMT'14 En-Fr), and deployed in a wide range of applications (from entity recognition for COVID-19 literature to large-scale online recommendation).

ORAL PAPERS

- [1] Model Tells You What to Discard: Adaptive KV Cache Compression for LLMs
Suyu Ge*, Yunan Zhang*, Liyuan Liu*, Minjia Zhang, Jiawei Han, and Jianfeng Gao
Proceedings of the Twelfth International Conference on Learning Representations (ICLR), 2024.
- [2] Bridging Discrete and Backpropagation: Straight-Through and Beyond.
Liyuan Liu, Chengyu Dong, Xiaodong Liu, Bin Yu, and Jianfeng Gao
Proceedings of the Thirty-seventh Annual Conference on Neural Information Processing Systems (NeurIPS), 2023.
- [3] Label Noise in Adversarial Training: A Novel Perspective to Study Robust Overfitting.
Chengyu Dong, Liyuan Liu, and Jingbo Shang
Proceedings of the Thirty-sixth Annual Conference on Neural Information Processing Systems (NeurIPS), 2022

- [4] Understanding the Difficulty of Training Transformers
Liyuan Liu, Xiaodong Liu, Jianfeng Gao, Weizhu Chen, and Jiawei Han
Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020.
- [5] Efficient Contextualized Representation: Compressing Language Models for Sequence Labeling
Liyuan Liu, Xiang Ren, Jingbo Shang, Jian Peng, and Jiawei Han
Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2018.
- [6] Heterogeneous Supervision for Relation Extraction: A Representation Learning Approach
Liyuan Liu*, Xiang Ren*, Qi Zhu, Shi Zhi, Huan Gui, Heng Ji, and Jiawei Han (* Equal Contribution)
Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017.
- [7] Community Detection Based on Structure and Content: A Content Propagation Perspective
Liyuan Liu, Linli Xu, Zhen Wang, and Enhong Chen
Proceedings of the 2015 International Conference on Data Mining (ICDM), 2015.

POSTER PAPERS, PREPRINTS, AND BLOGS

- [6] FlashRL: 8Bit Rollouts, Full Power RL
Liyuan Liu*, Feng Yao*, Dinghuai Zhang, Chengyu Dong, Jingbo Shang, Jianfeng Gao
<https://fengyao.notion.site/flash-rl>, 2025
- [7] Your Efficient RL Framework Secretly Brings You Off-Policy RL Training
Feng Yao*, Liyuan Liu*, Dinghuai Zhang, Chengyu Dong, Jingbo Shang, Jianfeng Gao
<https://fengyao.notion.site/off-policy-rl>, 2025
- [8] DenseMixer: Improving MoE Post-Training with Precise Router Gradient
Feng Yao^{*†}, Junxia Cui*, Ruohan Zhang*, Liyuan Liu[†], Shibo Hao, Li Zhang, Chengyu Dong, Shuohang Wang, Yelong Shen, Jianfeng Gao, Jingbo Shang
<https://fengyao.notion.site/moe-posttraining>, 2025
- [9] Decoder-Hybrid-Decoder Architecture for Efficient Reasoning with Long Generation
Liliang Ren, Congcong Chen, Haoran Xu, Young Jin Kim, Adam Atkinson, Zheng Zhan, Jiankai Sun, Baolin Peng, Liyuan Liu, Shuohang Wang, Hao Cheng, Jianfeng Gao, Weizhu Chen, Yelong Shen
arXiv preprint arXiv:2507.06607 [cs], 2025
- [10] Training Language Models to Generate Quality Code with Program Analysis Feedback
Feng Yao, Zilong Wang, Liyuan Liu, Junxia Cui, Li Zhong, Xiaohan Fu, Haohui Mai, Vish Krishnan, Jianfeng Gao, Jingbo Shang
arXiv preprint arXiv:2505.22704 [cs], 2025
- [11] Text Generation Beyond Discrete Token Sampling
Yufan Zhuang, Liyuan Liu, Chandan Singh, Jingbo Shang, Jianfeng Gao
arXiv preprint arXiv:2505.14827 [cs], 2025
- [12] Reinforcement Learning for Reasoning in Large Language Models with One Training Example
Yiping Wang, Qing Yang, Zhiyuan Zeng, Liliang Ren, Liyuan Liu, Baolin Peng, Hao Cheng, Xuehai He, Kuan Wang, Jianfeng Gao, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, Yelong Shen
arXiv preprint arXiv:2504.20571 [cs], 2025
- [13] Vector-ICL: In-context Learning with Continuous Vector Representations
Qingru Zhang, Chandan Singh, Liyuan Liu, Xiaodong Liu, Bin Yu, Jianfeng Gao, Tuo Zhao
Proceedings of the Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [14] LoRC: Low-Rank Compression for LLMs KV Cache with a Progressive Compression Strategy
Rongzhi Zhang, Kuan Wang, Liyuan Liu, Shuohang Wang, Hao Cheng, Chao Zhang, Yelong Shen
arXiv preprint arXiv:2410.03111 [cs], 2024
- [15] Grin: Gradient-Informed MoE

- Liyuan Liu, Young Jin Kim, Shuohang Wang, Chen Liang, Yelong Shen, Hao Cheng, Xiaodong Liu, Masahiro Tanaka, Xiaoxia Wu, Wenxiang Hu, Vishrav Chaudhary, Zeqi Lin, Chenruidong Zhang, Jilong Xue, Hany Awadalla, Jianfeng Gao, Weizhu Chen
arXiv preprint arXiv:2409.12136 [cs], 2024
- [16] Model Tells Itself Where to Attend: Faithfulness Meets Automatic Attention Steering
Qingru Zhang, Xiaodong Yu, Chandan Singh, Xiaodong Liu, Liyuan Liu, Jianfeng Gao, Tuo Zhao, Dan Roth, Hao Cheng
arXiv preprint arXiv:2409.10790 [cs], 2024
- [17] Learning a Decision Tree Algorithm with Transformers
Yufan Zhuang, Liyuan Liu, Chandan Singh, Jingbo Shang, Jianfeng Gao
Transactions on Machine Learning Research (TMLR), 2024.
- [18] Tell Your Model Where to Attend: Post-hoc Attention Steering for LLMs
Qingru Zhang, Chandan Singh, Liyuan Liu, Xiaodong Liu, Bin Yu, Jianfeng Gao, Tuo Zhao
Proceedings of the Twelfth International Conference on Learning Representations (ICLR), 2024.
- [19] Fast-ELECTRA for Efficient Pre-training
Chengyu Dong, Liyuan Liu, Hao Cheng, Jingbo Shang, Jianfeng Gao, Xiaodong Liu
Proceedings of the Twelfth International Conference on Learning Representations (ICLR), 2024.
- [20] Toward Student-oriented Teacher Network Training for Knowledge Distillation
Chengyu Dong, Liyuan Liu, Jingbo Shang
Proceedings of the Twelfth International Conference on Learning Representations (ICLR), 2024.
- [21] Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone
Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, Alon Benhaim, Misha Bilenko, Johan Bjorck, Sébastien Bubeck, Martin Cai, Qin Cai, Vishrav Chaudhary, Dong Chen, Dongdong Chen, Weizhu Chen, Yen-Chun Chen, Yi-Ling Chen, Hao Cheng, Parul Chopra, Xiyang Dai, Matthew Dixon, Ronen Eldan, Victor Fragoso, Jianfeng Gao, Mei Gao, Min Gao, Amit Garg, Allie Del Giorno, Abhishek Goswami, Suriya Gunasekar, Emman Haider, Junheng Hao, Russell J. Hewett, Wenxiang Hu, Jamie Huynh, Dan Iter, Sam Ade Jacobs, Mojan Javaheripi, Xin Jin, Nikos Karampatziakis, Piero Kauffmann, Mahoud Khademi, Dongwoo Kim, Young Jin Kim, Lev Kurilenko, James R. Lee, Yin Tat Lee, Yanzhi Li, Yunsheng Li, Chen Liang, Lars Liden, Xihui Lin, Zeqi Lin, Ce Liu, Liyuan Liu, Mengchen Liu, Weishung Liu, Xiaodong Liu, Chong Luo, Piyush Madan, Ali Mahmoudzadeh, David Majercak, Matt Mazzola, Caio César Teodoro Mendes, Arindam Mitra, Hardik Modi, Anh Nguyen, Brandon Norick, Barun Patra, Daniel Perez-Becker, Thomas Portet, Reid Pryzant, Heyang Qin, Marko Radmilac, Liliang Ren, Gustavo de Rosa, Corby Rosset, Sambudha Roy, Olatunji Ruwase, Olli Saarikivi, Amin Saied, Adil Salim, Michael Santacrose, Shital Shah, Ning Shang, Hiteshi Sharma, Yelong Shen, Swadheen Shukla, Xia Song, Masahiro Tanaka, Andrea Tupini, Praneetha Vaddamanu, Chunyu Wang, Guanhua Wang, Lijuan Wang, Shuohang Wang, Xin Wang, Yu Wang, Rachel Ward, Wen Wen, Philipp Witte, Haiping Wu, Xiaoxia Wu, Michael Wyatt, Bin Xiao, Can Xu, Jiahang Xu, Weijian Xu, Jilong Xue, Sonali Yadav, Fan Yang, Jianwei Yang, Yifan Yang, Ziyi Yang, Donghan Yu, Lu Yuan, Chenruidong Zhang, Cyril Zhang, Jianwen Zhang, Li Lyna Zhang, Yi Zhang, Yue Zhang, Yunan Zhang, Xiren Zhou
arXiv preprint arXiv:2404.14219 [cs], 2024
- [22] Sparse Backpropagation for MoE Training
Liyuan Liu, Jianfeng Gao, Weizhu Chen
arXiv preprint arXiv:2310.00811 [cs], 2023
- [23] Understand and Modularize Generator Optimization in Electra-style Pretraining
Chengyu Dong, Liyuan Liu, Hao Cheng, Jingbo Shang, Jianfeng Gao, Xiaodong Liu
Proceedings of the 40th International Conference on Machine Learning (ICML), 2023.
- [24] Piled: An identify-and-localize framework for few-shot event detection
Sha Li, Liyuan Liu, Yiqing Xie, Heng Ji, Jiawei Han
arXiv preprint arXiv:2202.07615 [cs], 2022
- [25] Ucphrase: Unsupervised context-aware quality phrase tagging
Xiaotao Gu, Zihan Wang, Zhenyu Bi, Yu Meng, Liyuan Liu, Jiawei Han, Jingbo Shang
Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data mining (KDD), 2021

- [26] Multi-head or Single-head? An Empirical Comparison for Transformer Training
Liyuan Liu, Jialu Liu, and Jiawei Han
arXiv preprint arXiv:2106.09650 [cs], 2021
- [27] Data Quality Matters For Adversarial Training: An Empirical Study
Chengyu Dong, Liyuan Liu, and Jiawei Han
arXiv preprint arXiv:2102.07437 [cs], 2021
- [28] On the Transformer Growth for Progressive BERT Training
Xiaotao Gu, Liyuan Liu, Hongkun Yu, Jing Li, Chen Chen, and Jiawei Han
Proceedings of the 2021 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), 2021.
- [29] Empower Distantly Supervised Relation Extraction with Collaborative Adversarial Training
Tao Chen, Haochen Shi, Liyuan Liu, Siliang Tang, Jian Shao, Zhigang Chen, and Yueting Zhuang
Proceedings of the 2021 AAAI Conference on Artificial Intelligence (AAAI), 2021.
- [30] Graph Clustering with Embedding Propagation
Carl Yang, Liyuan Liu, Mengxiong Liu, Zongyi Wang, Chao Zhang, and Jiawei Han
Proceedings of the 2020 IEEE International Conference on Big Data (BigData), 2020.
- [31] Towards Adaptive Residual Network Training: A Neural-ODE Perspective
Chengyu Dong, Liyuan Liu, Zichao Li, and Jingbo Shang
Proceedings of the 2020 International Conference on Machine Learning (ICML), 2020.
- [32] Joint Aspect-Sentiment Analysis with Minimal User Guidance
Honglei Zhuang, Fang Guo, Chao Zhang, Liyuan Liu, and Jiawei Han
Proceedings of the 2020 ACM SIGIR International Conference on Research and development in Information Retrieval (SIGIR), 2020.
- [33] Facet-Aware Evaluation for Extractive Summarization
Yuning Mao, Liyuan Liu, Qi Zhu, Xiang Ren, and Jiawei Han
Proceedings of the 2020 Annual Conference of the Association for Computational Linguistics (ACL), 2020.
- [34] Learning to Contextually Aggregate Multi-Source Supervision for Sequence Labeling
Ouyu Lan, Xiao Huang, Bill Yuchen Lin, He Jiang, Liyuan Liu, and Xiang Ren
Proceedings of the 2020 Annual Conference of the Association for Computational Linguistics (ACL), 2020.
- [35] On the Variance of the Adaptive Learning Rate and Beyond
Liyuan Liu, Haoming Jiang, Pengcheng He, Weizhu Chen, Xiaodong Liu, Jianfeng Gao, and Jiawei Han
Proceedings of the 2020 International Conference on Learning Representations (ICLR), 2020.
- [36] NetTaxo: Automated Topic Taxonomy Construction from Large-Scale Text-Rich Network
Jingbo Shang, Xinyang Zhang, Liyuan Liu, Sha Li, and Jiawei Han
Proceedings of the 2020 International World Wide Web Conference (WWW), 2020.
- [37] Partially-Typed NER Datasets Integration: Connecting Practice to Theory
Shi Zhi*, Liyuan Liu*, Yu Zhang, Shiyin Wang, Qi Li, Chao Zhang, and Jiawei Han
arXiv preprint arXiv:2005.00502 [cs], 2020.
- [38] Very Deep Transformers for Neural Machine Translation
Xiaodong Liu, Kevin Duh, Liyuan Liu, and Jianfeng Gao
arXiv preprint arXiv:2008.07772 [cs], 2020.
- [39] CrossWeigh: Training Named Entity Tagger from Imperfect Annotations
Zihan Wang*, Jingbo Shang*, Liyuan Liu*, Lihao Lu, Jiacheng Liu, and Jiawei Han (* Equal Contribution)
Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2019.
- [40] Looking Beyond Label Noise: Shifted Label Distribution Matters in Distantly Supervised Relation Extraction
Qinyuan Ye*, Liyuan Liu*, Maosen Zhang, and Xiang Ren (* Equal Contribution)

Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2019.

- [41] Arabic Named Entity Recognition: What Works and What's Next
Liyuan Liu, Jingbo Shang, and Jiawei Han
Proceedings of the 2019 Arabic Natural Language Processing Workshop (WANLP), co-located with ACL 2019, 2019.
- [42] Reliability-aware Dynamic Feature Composition for Name Tagging
Ying Lin, Liyuan Liu, Heng Ji, Dong Yu, and Jiawei Han
Proceedings of the 2019 Annual Meeting of the Association for Computational Linguistics (ACL), 2019.
- [43] Cross-relation Cross-bag Attention for Distantly-supervised Relation Extraction
Yujin Yuan, Liyuan Liu, Siliang Tang, Zhongfei Zhang, Yueting Zhuang, Fei Wu, and Xiang Ren
Proceedings of the 2019 AAAI Conference on Artificial Intelligence (AAAI), 2019.
- [44] Raw-to-End Name Entity Recognition in Social Media
Liyuan Liu, Zihan Wang, Jingbo Shang, Dandong Yin, Heng Ji, Xiang Ren, Shaowen Wang, and Jiawei Han
arXiv preprint arXiv:1908.05344 [cs], 2019.
- [45] Learning Named Entity Tagger using Domain-Specific Dictionary
Jingbo Shang*, Liyuan Liu*, Xiaotao Gu, Xiang Ren, Teng Ren, and Jiawei Han (* Equal Contribution)
Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2018.
- [46] Empower Sequence Labeling with Task-Aware Neural Language Model
Liyuan Liu, Jingbo Shang, Frank Xu, Xiang Ren, Huan Gui, Jian Peng, and Jiawei Han
Proceedings of the 2018 AAAI Conference on Artificial Intelligence (AAAI), 2018.
- [47] TrioVecEvent: Embedding-Based Online Local Event Detection in Geo-Tagged Tweet Streams
Chao Zhang, Liyuan Liu, Dongming Lei, Quan Yuan, Honglei Zhuang, Tim Hanratty, and Jiawei Han
Proceedings of the 2017 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2017.
- [48] Wikidata Vandalism Detection—The Loganberry Vandalism Detector at WSDM Cup 2017
Qi Zhu, Hongwei Ng, Liyuan Liu, Ziwei Ji, Bingjie Jiang, Jiaming Shen, and Huan Gui
Proceedings of the WSDM Cup 2017 Notebook Papers (WSDM Cup), 2017,